

Lemma. Let V be a finite dimensional vector space over F .

If the characteristic polynomial of a linear operator $T: V \rightarrow V$ splits over F , then there exists a basis β s.t. $[T]_\beta$ is upper-triangular matrix.

Proof. put $n = \dim V$. Using mathematical induction for n . If $n=1$, trivial

Suppose that for $n > 1$, the statement holds for every spaces having dimension as $n-1$.

Let V be a vector space with $\dim V = n$, and let $T: V \rightarrow V$ be a linear.

Since the characteristic polynomial of T splits over F , by assumption,

there exists a eigenvalue $\lambda_1 \in F$ corresponding to $v_1 \in V$.

By the Replacement Thm, we obtain a basis for V containing v_1

$$\{v_1, v_2, \dots, v_n\}$$

Denote $W_1 = \langle v_1 \rangle$, $W_2 = \langle v_2, \dots, v_n \rangle$. Clearly, $V = W_1 \oplus W_2$.

Observe that W_1 is T -invariant but W_2 is not. Define two projections

$$P_1: V \rightarrow V: x_1 + x_2 \mapsto x_1 \quad \text{and} \quad P_2: V \rightarrow V: x_1 + x_2 \mapsto x_2, \quad (x_1 \in W_1, x_2 \in W_2)$$

Then, for any $x \in W_2$, $(P_2 \circ T)(x) \in W_2$, thus we can consider a restriction

$$P_2 \circ T|_{W_2}: W_2 \rightarrow W_2: x \mapsto (P_2 \circ T)(x).$$

By the assumption, there exists a basis $\{w_2, \dots, w_n\}$ for W_2 s.t. for $2 \leq i \leq n$,

$$(P_2 \circ T)(w_i) = \sum_{j=2}^i a_{ij} w_j \quad \text{for some } a_{ij} \in F.$$

Now, define for $1 \leq k \leq n-1$, $u_{k+1} = v_1 + w_{k+1}$. Then,

$\{v_1, u_2, \dots, u_n\}$ is a basis for V , and satisfies for all $1 \leq i \leq n$

$$T(v_1) = \lambda_1 v_1$$

$$T(u_2) = T(v_1 + w_2) = T(v_1) + T(w_2) = \lambda_1 v_1 + a_{22} w_2$$

$$T(u_3) = T(v_1 + w_2 + w_3) = \lambda_1 v_1 + a_{22} w_2 + a_{33} w_3$$

⋮

$$T(u_n) = \lambda_1 v_1 + a_{22} w_2 + \dots + a_{nn} w_n.$$

